



**COMSATS University Islamabad,
Sahiwal Campus.**

E-Clinic

A Project Presented To

**COMSATS University, Islamabad
Sahiwal Campus**

In partial fulfillment

of the requirement for the degree of

Bachelor of Science in Computer Science (2021-2025)

By

Taha Nadeem

CIIT/FA21-BCS-021/SWL

Muhammad Ahmed

CIIT/FA21-BCS-049/SWL

DECLARATION

We hereby declare that this system, neither whole nor as a part, has been copied out from any source. It is further declared that we have developed this website and accompanied report entirely based on our personal efforts. If any part of this project is proved to be copied out from any source or found to be reproduction of some other. We will stand by the consequences. No Portion of the work presented has been submitted of any application for any other degree or qualification of this or any other university or institute of learning.

Taha Nadeem

Muhammad Ahmed

CERTIFICATE OF APPROVAL

It is to certify that the final year project of BS (CS) “E clinic” was developed by **Taha Nadeem (FA21-BCS-021)** and **Muhammad Ahmed (FA21-BCS-049)** under the supervision of “**Mr. Ali Sher Kashif**” and that in their opinion; it is fully adequate, in scope and quality for the degree of Bachelor of Science in Computer Sciences.

Supervisor

Mr. Ali Sher Kashif

Lecturer

Department of Computer Science

COMSATS University Islamabad, Sahiwal Campus

External Examiner

Dr. Zafar Iqbal

Head of Department

Department of Computer Science

COMSATS University Islamabad, Sahiwal Campus

Executive Summary

E Clinic is an innovative and comprehensive healthcare platform designed to transform and streamline the patient–doctor experience by combining modern online features with traditional in-person consultations, ultimately addressing the challenges and inefficiencies of conventional healthcare systems. The platform enables patients and doctors to register securely through a robust authentication system supported by JWT protocols, ensuring data protection and privacy compliance. Patients can conveniently book appointments for either physical visits or secure video consultations, browse a detailed directory of doctors filtered by specialization, experience, ratings, and location, and make more informed healthcare decisions. Doctors are empowered with tools to manage their profiles, appointment schedules, patient histories, and even handle payments, while being able to issue digital prescriptions and view uploaded lab reports in a secure, streamlined environment. Built using modern web technologies such as Node.js, Prisma, PostgreSQL, Vue.js, and styled through contemporary front-end practices, E Clinic reflects current industry standards for high-performance web applications. The project followed an agile methodology with iterative development and frequent user feedback to ensure continuous improvement, supported by tools like Docker for scalable deployment, version control practices, and microservices architecture for modular, maintainable code. Manual and system testing validated the reliability and usability of the platform, covering functional requirements like user authentication, appointment management, prescription handling, lab report management, and integrated online payment systems that accept credit cards, debit cards, and digital wallets. The user interface was designed with human-computer interaction principles to ensure a clear, intuitive, and responsive experience across devices, while maintaining consistent performance and security standards. Additionally, the platform encourages transparency through a doctor’s rating and review feature, allowing patients to share their experiences and helping others choose the right care providers with confidence. Beyond its current features, E Clinic envisions future enhancements like machine learning–based symptom checkers and diagnostic tools to predict illnesses and suggest treatments, thereby improving early detection and personalizing patient care even further. Overall, E Clinic provides an efficient, secure, and user-centered approach to healthcare delivery, bridging critical gaps in accessibility, trust, and convenience, and setting a foundation for a smarter and more connected healthcare future for patients, doctors, and administrators alike.

Acknowledgement

All praise is to Almighty Allah who bestowed upon us a minute portion of His boundless knowledge by virtue of which we were able to accomplish this challenging task. We are greatly indebted to our project supervisor “**Mr. Ali Sher Kashif**”. Without their personal supervision, advice and valuable guidance, completion of this project would have been doubtful. We are deeply indebted to them for their encouragement and continual help during this work.

And we are also thankful to our parents and family who have been a constant source of encouragement for us and brought us the values of honesty & hard work.

Taha Nadeem

Muhammad Ahmed

Abbreviations

SRS	Software Require Specification
UI	User Interface
UX	User Experience
UAT	User Acceptance Testing
CI	Continuous Integration

Table of Contents

1.	Introduction	1
1.1	Modules.....	2
1.2	Relevance to Course Modules	5
1.3	Project Background	6
1.4	Literature Review	6
1.5	Methodology.....	7
1.6	Software Development Lifecycle	8
2	Problem Definition.....	10
2.1	Problem Statement.....	10
2.2	Deliverables and Development Requirements	10
2.2.1	User Friendly.....	10
2.2.2	User Authentication:.....	10
2.2.3	Appointment Booking Functionality.....	10
2.2.4	Doctor Search and Verification:.....	10
2.2.5	Prescription and Lab Report Access	10
2.2.6	Online Payment Integration.....	11
2.2.7	Front-end Development.....	11
2.2.8	Database Management	11
2.2.9	Integration of Third-Party APIs	11
2.2.10	Security Measures	11
2.2.11	Testing and Quality Assurance	11
2.2.12	Deployment and Scalability	11
3	Requirement Analysis	13
3.1	Use Case Diagram	13
3.2	Functional Requirements.....	30
3.2.1	Doctor Module	31
3.2.2	Patient Module	31
3.2.3	User Authentication Module.....	32
3.2.4	Admin Module	32
3.3	Non-Functional Requirements.....	32
3.3.1	Security	33
3.3.2	Performance	33
3.3.3	Reliability.....	33

3.3.4	Usability.....	34
4	Design and Architecture.....	36
4.1	System Architecture.....	36
4.2.	Data Representation.....	37
4.3	Process Flow	39
4.4	Design Models	41
5	Implementation.....	44
5.1	User Interface	45
6	Testing and Evaluation.....	50
6.1	Manual Testing.....	50
6.2.	System Testing	50
7.	Conclusion.....	53
7.1	Future Work	53
8.	References	55

List of Figures

Figure 1.1 Agile Methodology	8
Figure 3.1 Use-case Patient and Doctor Registration.....	13
Figure 3.2 Use-case Patient Appointment Booking	14
Figure 3.3 Use-case Patient Appointment Cancel, Accept or Reject	14
Figure 3.4 Use-Case Online Payment	15
Figure 3.5 Use-case Review and Ratings of Doctor.....	16
Figure 4.1 System Architecture of E clinic	36
Figure 4.2 Complete Process Flow of E clinic	40
Figure 4.3 Class Diagram of E clinic	41
Figure 4.4 Sequence Diagram of E clinic.....	42
Figure 5.1 Login/Signup of E clinic.....	46
Figure 5.2 Feature of E clinic	47
Figure 5.3 Dashboard for Patients	47
Figure 5.4 Doctors Page of E clinic.....	48
Figure 5.5 About Us Page of E clinic.....	48

List of Tables

Table 3.1 Use-Case Patient and Doctor Registration	16
Table 3.2 Use-Case Patient Appointment Booking.....	19
Table 3.3 Use-Case Patient Appointment Cancel, Accept or Reject.....	22
Table 3.4 Use-Case Online Payment.....	24
Table 3.5 Use-Case Review and Ratings of Doctor	27
Table 4.1 Data Representation of E clinic.....	37
Table 6.1 Manual Testing of E clinic	50
Table 6.2 System Testing of E clinic.....	51

Chapter 1

Introduction

1. Introduction

In today's fast-paced world, the healthcare industry has witnessed significant advancements in technology, leading to the emergence of innovative solutions that simplify and streamline healthcare processes. One such groundbreaking solution is E clinic, a digital platform that revolutionizes the way patients interact with healthcare providers. This project aims to develop a system that incorporates essential features such as login/signup, appointment booking (both physically and online), doctor verification, prescription management, lab report access, and online payment options, cancellation, accept and then prescribe to them. Protecting patient data and ensuring privacy compliance was a concern. Ensuring that the data entered in E clinic is accurate and of high quality is essential for effective patient care. Human errors or inconsistent data entry can lead to problems.

E clinic system may incorporate patient portals that empower patients to access their medical records, view test results, schedule appointments, and communicate with healthcare providers securely online. Patient engagement is essential for better healthcare outcomes. We are making websites as well as applications by using different languages like node.js, Prisma, PostgreSQL, Vue.

One of the main modules is the patient's module. In which patients can see a list of messages of doctors with their availability where you communicate with them for his/her concerned problem about health. Also, he or she can view all the doctors of different professions. Patients can provide a review of their experiences. There would be the option of previous and upcoming appointments. Online consultation with time of appointment. Another module is the doctor's module where doctors firstly sign up and log in to their account. Where all his personal information like email, address, CNIC, profession, concerned hospital, experience.

In case doctors forget their password there is an option to set it via email or SMS. There is a sub module which is a payment module in which all information is present of pending payments, total amount. Appointment management where he /she cancels or accepts the upcoming appointment. Doctors will check all the information of the patient and his medical history, notifications, account settings, payment methods. E clinic systems often include practice management software, which assists healthcare providers in managing appointments, billing, and administrative tasks efficiently. The record of a specific patient only shows to the specific doctor i.e., his all history including his tests, medicines. We have explored other e-commerce sites related to our project which includes dawai.pk (Kidwai, 2013), marham.pk (), oladoc.com (Zuberi, 2017).

Each patient will be his own account. There is also a feedback option for patients to give his reviews about his/her experience. First, he must sign up, afterwards he will login his account then finds the doctor according to medical conditions to appoint online, after that he would provide his medical history. Laboratory tests, medical records, pharmacy, recommendation of medicines related to his health problem. As well as not only E clinic website but also its an android app which will work same as the website including the appointment, prescription feature, different doctors related to different health issues like cardio, gastroenterologist, psychiatrist etc. on the other side, same as patient accounts and his history.

The government of Pakistan may have introduced initiatives or regulations to encourage the adoption of E clinic systems and ensure that healthcare providers comply with certain standards. By leveraging the power of technology, this E clinic system will enhance the efficiency, accessibility, and convenience of healthcare services for both patients and doctors.

1.1 Modules

Modules are distinct components or phases of a project. Breaking down a project into modules helps in planning, organizing, and managing different aspects efficiently.

1.1.1 Doctor Module

The Doctor module has the following possible functionalities:

i. Signup

This module allows doctors to create an account on the platform. Doctors provide their personal information, credentials, contact details, and experience during the signup process.

ii. Dashboard

The dashboard serves as the central hub for doctors to access various features and information. It provides an overview of the doctor's schedule, upcoming appointments, and notifications.

iii. Appointment Management

Doctors can view, edit, and manage their appointment schedule through this module. They can see a list of all their appointments, both upcoming and past. Rescheduling or canceling appointments can be done here, with notifications sent to patients.

iv. New Appointment Requests

When a patient requests an appointment, it appears in this module for the doctor's review. Doctors can accept or reject appointment requests based on their availability.

v. Upcoming Requests

This section displays a list of approved upcoming appointments. Doctors can access patient details, medical history, and appointment notes to prepare for the visit.

vi. Completed Appointments

Once an appointment is concluded, it moves to the "Completed Appointments" section. Doctors can review the appointment details, notes, and patient information. This module also allows doctors to add prescriptions or treatment plans for the patient.

vii. Add prescription for completed Appointment

Enables doctors to create and store prescriptions for patients. Doctors can specify medications, dosages, and instructions for the patient. Prescription data should be securely recorded for future reference.

viii. Availability Management

Allows doctors to set their availability for appointments. Doctors can specify their working hours, days off, and vacation periods. Ensures that appointment requests align with the doctor's schedule.

ix. Charges Management

It includes the about the payment that how much payment is pending, how much total amount is required.

1.1.2 Settings

The settings module has the following possible functionalities:

i. Email

This setting allows doctors to specify and update their email address associated with their account. It's essential for communication, notifications, and account recovery.

ii. Phone

Doctors can input their contact phone number, which can be used for appointment reminders, emergency notifications, and communication with patients and hospital staff.

iii. Password

This setting enables doctors to change their login password securely. Regular password updates enhance account security.

iv. Hospital Settings

Hospital settings include the information about the hospital of doctor i.e., hospital name address, city, state.

1.1.3 Payments

The payments module has the following possible functionalities:

i. Payment Methods:

Patients should be able to choose from various payment methods such as credit/debit cards, electronic wallets, or even insurance claims.

ii. Payment Processing:

Integration with payment gateways to securely process payments made by patients.

iii. Payment Notifications:

Automated notifications to inform patients about payment due dates, successful payments, and payment receipts.

iv. Withdrawal

The doctor should have the capability to request withdrawals for the funds they have earned through patient payments.

v. Transactions History

A detailed record of all financial transactions related to the doctor's services, including payments received, withdrawals made, and any adjustments or refunds.

1.1.4 Patient Module

The Patient module have the following possible functionalities:

i. Doctor Search

Search Functionality: Patients can search for doctors based on various criteria such as specialty, location, availability, and patient reviews.

Filtering and Sorting: Patients can refine search results through filters like gender, ratings, and languages spoken.

Doctor Profiles: Detailed profiles for each doctor, including their qualifications, experience, specialties, working hours, and contact information.

ii. Appointment Booking

Scheduling: Patients can request appointments by selecting a preferred date and time slot based on the doctor's availability.

Confirmation: Patients receive confirmation of their appointment along with details via email or SMS.

Rescheduling and Cancellation: Patients can reschedule or cancel appointments with appropriate notice.

iii. Appointments Management

Upcoming Appointments: Patients can view a list of their upcoming appointments, including date, time, and location.

iv. Previous Appointments

Patients can view a list of previous appointments, including date, time, and location.

v. Appointment Reminders

Automated reminders via email or SMS to ensure patients don't miss their appointments.

History: Access to the patient's appointment history for reference.

vi. Payments Management

Payment Methods: Patients can securely add and manage payment methods such as credit cards

Payment History: Patients can review their payment history, including past transactions and outstanding balances.

vii. Access Prescription

Patients can check prescription with specific date at which doctor gave him/her with his name, CNIC.

1.2 Relevance to Course Modules

E clinic web-based project directly reflects the theoretical and practical knowledge gained throughout the Computer Science curriculum. It integrates multiple core modules, including:

i. Web Development & Programming Fundamentals:

The project applies HTML, CSS, JavaScript to build interactive, user-friendly interfaces and server-side logic. This aligns with the programming and frontend-backend development topics covered in web and application development courses.

ii. Software Engineering:

Through documentation like the SRS (Software Requirements Specification), SDD (Software Design Document), and modular design, the project demonstrates key principles from software engineering, such as system analysis, design patterns, version control, and collaborative development.

iii. Database Management Systems (DBMS):

The backend connects to a database to manage patients, doctors, and appointments, reinforcing practical DBMS skills like CRUD operations, data modeling, and real-time data handling.

iv. Human-Computer Interaction (HCI):

The design of the user interface for patients and administrators reflects HCI principles like usability, navigation, and responsive design.

v. Computer Networks & Deployment:

The use of Docker and cloud-ready configurations ties in with networking and deployment concepts, showcasing how real-world applications are hosted, scaled, and maintained in production environments.

vi. Project Management & Documentation:

Planning, progress tracking, and report writing demonstrate project management skills that are critical in real-world software development cycles.

1.3 Project Background

The E clinic project is initiated to address the growing demand for a comprehensive healthcare management system that incorporates digital features to improve patient care and enhance doctor-patient interactions. Traditional healthcare systems often face challenges such as long waiting times, limited accessibility, and inefficient record management. By developing an E clinic system, these issues can be mitigated, leading to a more patient-centric and efficient healthcare experience.

1.4 Literature Review

We have explored different research papers and different sites to check up on the current condition of online health applications. Such a review can provide insights into the current state of E clinic systems, their benefits, challenges, and emerging trends. Explore the historical development of E clinic systems and their evolution over time. Key milestones and technological advancements that have contributed to the growth of these systems. I've read research paper about Healthcare System.

Where country has been facing many challenges like the shortage of trained human and financial resources with Poor Governance, political interference, and lack of accountability for poor performance etc. In Pakistan, "Health system has been designed by politicians, while it has been implemented by the health professionals on the ground level" (Shaikh & Rabbani, 2004). Especially the access and stability of the country's rural population regarding health services is a major problem "because of severe shortage of healthcare professionals and inadequate allocation of funds for Primary Healthcare sector" (Punjani et al. 2014). (Hashami, 2020)

The Dawai.pk website (Kidwai, 2013) gives us so much information One of the key concerns in online pharmacies is the verification of prescriptions. Dawai.com (Kidwai, 2013) and similar platforms need to establish robust systems to verify the authenticity of prescriptions to ensure that only authorized medications are dispensed. But in our system, the doctor itself gives the prescription to patient addresses to his/her health issue, and his medical history. There is a proper talk between doctor and patient likewise also in our app.

While Marham.pk (Imam, 2016) may offer a range of services, it may not cover all healthcare needs comprehensively. Certain specialized medical services or procedures may not be available through the platform. But we have proper features of prescriptions, lab reports, pharmacy. Moreover, everything will be saved on our site, our history of medicines, prescriptions, reports.

The government has consistently put in the effort to reduce healthcare costs in hospitals and has established autonomous bodies to relegate a more efficient and effective healthcare delivery. Pakistan has a long way ahead toward the development of an effective, accessible, and affordable healthcare system. The system is plagued with numerous flaws, ranging from inadequate infrastructure to inequitable distribution of healthcare facilities. The lack of adequate healthcare infrastructure is one of Pakistan's biggest challenges. There is an extreme shortage of healthcare facilities, including hospitals, clinics, and diagnostic centers (Hashami, 2020)

A healthcare system is one of the most essential pillars of any country. The primary role of a healthcare system is to ensure that all people get the best available health facilities in a timely, acceptable, affordable, and accessible manner. However, for a healthcare system to function as such, it requires proper infrastructure and financial support (Muhammad Q, 2023).

1.5 Methodology

The development of the E clinic system will follow an iterative and agile software development approach. The project team will employ user-centered design principles, starting with requirements gathering and analysis. This will be followed by the design and prototyping phase, where wireframes and user interface designs will be created. The system will be developed using appropriate programming languages and frameworks, with regular testing and feedback cycles to ensure quality and functionality. The implementation will be accompanied by thorough documentation and user training to ensure seamless adoption and utilization of the E clinic system. Agile methodology is an iterative and flexible approach to software development and project management. Key principles include iterative development, collaboration, customer involvement, adaptive planning, cross-functional teams, continuous integration and testing, regular reflection, and improvement. Common frameworks within Agile include Scrum, Kanban, and Extreme Programming, each with specific practices and roles. The methodology prioritizes adaptability, customer satisfaction, and open communication to deliver valuable software incrementally. As Shown in Figure 1.1.



Figure 1.1 Agile Methodology

1.6 Software Development Lifecycle

The E clinic system will adhere to a structured SDLC (Software Development Life Cycle) to ensure successful project execution. The chosen SDLC (Software Development Life Cycle) model may be an iterative approach, such as Agile or Scrum, which allows for flexibility and continuous improvement throughout the development process. This model enables incremental development, frequent feedback, and rapid iterations, ultimately resulting in a robust and user-friendly E clinic system. The project team will establish milestones, conduct regular meetings, and maintain effective communication channels to ensure timely delivery and stakeholder satisfaction.

Chapter 2

Problem Definition

2 Problem Definition

Healthcare services in hospitals in Pakistan vary between public and private facilities. Public hospitals, funded by the government, aim to provide services to a broad population but may face challenges like overcrowding. Private hospitals in urban areas generally offer higher standards of care for those who can afford it. Both types of hospitals provide a range of services, including emergency care, surgeries, and specialized treatments. Challenges include shortages of healthcare professionals, inadequate infrastructure, and disparities between urban and rural areas. They do not have any proper software to keep record of Lab Reports and Prescriptions of the patients. Our project can be defined with the help of following statements:

2.1 Problem Statement

The current healthcare system lacks an efficient and user-friendly platform for managing clinic appointments, doctor consultations, and medical records. Patients face difficulties in finding suitable doctors, booking appointments physically and online, accessing their prescription and lab reports, and making payments. This results in inconvenience, time wastage, and potential errors in the healthcare process. Therefore, there is a need for an E clinic solution that addresses these issues and provides a seamless experience for patients, doctors, and clinic administrators.

2.2 Deliverables and Development Requirements

Deliverables and Development Requirements for the E clinic Project:

2.2.1 User Friendly: Develop an intuitive web-based interface that allows users to easily navigate and access the various features of the E clinic platform.

2.2.2 User Authentication: Implement a secure login and signup system for patients and doctors, ensuring the confidentiality of their personal information.

2.2.3 Appointment Booking Functionality: Enable patients to book appointments with doctors of their choice, either physically or online, based on availability. Provide real-time updates on appointment status and notifications for both patients and doctors.

2.2.4 Doctor Search and Verification: Integrate a search functionality that enables patients to find doctors based on specialty, location, availability, and user reviews. Implement a verification process to ensure the credibility and legitimacy of doctors listed on the platform.

2.2.5 Prescription and Lab Report Access: Develop a feature that allows doctors to digitally generate and store prescriptions for patients. Enable patients to access their prescriptions and lab reports securely through the E clinic platform.

2.2.6 Online Payment Integration: Integrate a secure online payment system that enables patients to make payments for appointments, prescriptions, and other healthcare services conveniently and safely.

2.2.7 Front-end Development: Utilize modern web development technologies such as Vue.js, Nuxt.js and JavaScript to create an interactive and responsive user interface.

Back-end Development: Build a robust server-side application using a suitable programming language and a web framework (e.g., Node Js) to handle data storage, user authentication, appointment management, and other core functionalities.

2.2.8 Database Management: Design and implement a database schema to store user information, appointment details, prescriptions, lab reports, and other relevant data. Utilize a reliable database management system (e.g., MySQL, PostgreSQL, or MongoDB) to ensure efficient data storage and retrieval.

2.2.9 Integration of Third-Party APIs: Utilize APIs provided by payment gateways for secure online payment integration. Implement APIs for doctor search, verification, and appointment scheduling if available.

2.2.10 Security Measures: Implement appropriate security measures such as data encryption, secure user authentication, and protection against common web vulnerabilities (e.g., cross-site scripting, SQL injection) to ensure the confidentiality and integrity of user data.

2.2.11 Testing and Quality Assurance: Conduct thorough testing, including unit testing, integration testing, and user acceptance testing, to ensure the reliability, functionality, and usability of the E clinic platform.

2.2.12 Deployment and Scalability: Deploy the application on a reliable hosting platform or cloud infrastructure to ensure accessibility and scalability as the user base grows. Consider factors such as server performance, load balancing, and backup and recovery mechanisms.

By addressing these problem statements and implementing the specified deliverables and development requirements, the E clinic solution aims to streamline the healthcare process, enhance patient-doctor interactions, and provide a secure and convenient platform for managing clinic appointments, medical records, and online payments.

Chapter 3

Requirement Analysis

3 Requirement Analysis

Our system requirements are registration of doctor and patient, appointment booking, online payments, prescriptions, Lab reports and reviews. We will use Use-Case as a requirement analysis technique that describes the interactions between a user (also known as an actor) and a system to achieve a specific goal. Use cases are a useful tool for identifying and organizing the functional requirements of a system.

3.1 Use Case Diagram

A use case diagram is a visual representation in the Unified Modeling Language (UML) that illustrates the interactions between different actors (users or systems) and a system or software application. It provides a high-level view of how the system or software behaves from a user's perspective. Use case diagrams are particularly useful for understanding the functional requirements of a system and its various use cases or scenarios. As shown in Figure 3.1.

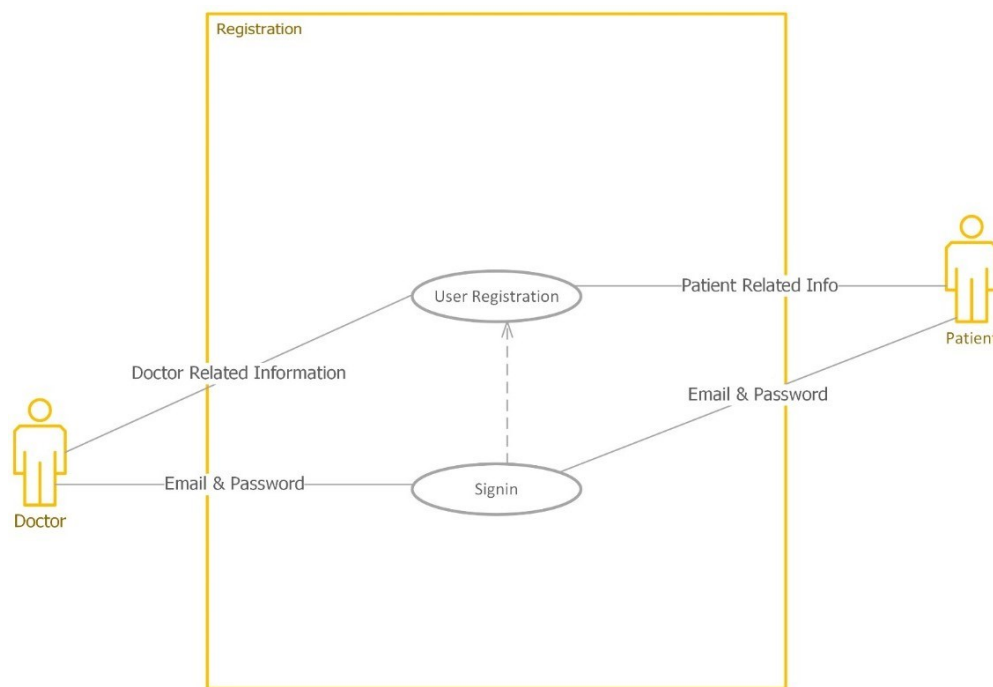


Figure 3.1 Use-case Patient and Doctor Registration

These two images represent the working of Appointment booking. In a patient appointment booking system, patients can book appointments by selecting a date and time. They can cancel

appointments within 24 hours of booking. Doctors have the authority to accept or reject appointment requests. The system manages these interactions, and a use case diagram visually represents these key functionalities, involving patients, doctors, and the system. As shown in Figure 3.2 and Figure 3.3.

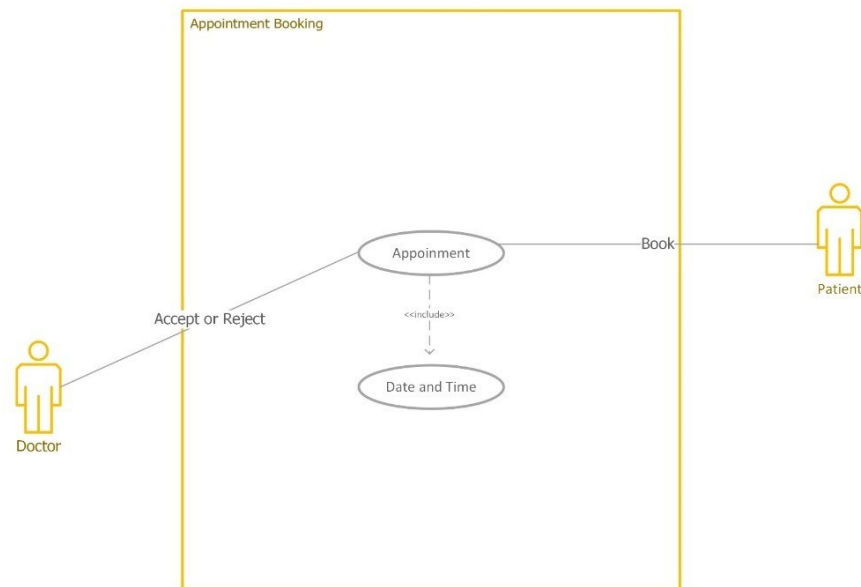


Figure 3.2 Use-case Patient Appointment Booking

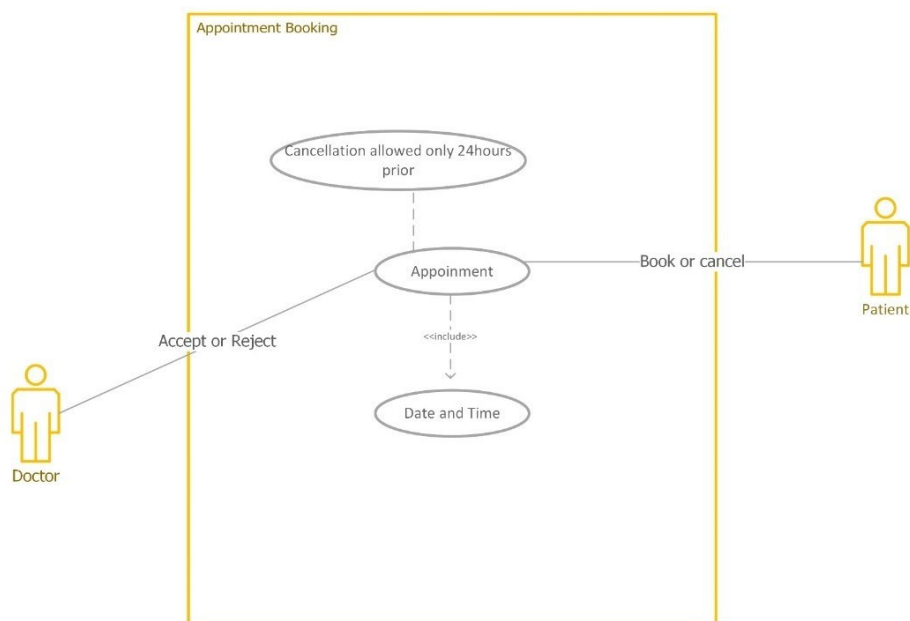


Figure 3.3 Use-case Patient Appointment Cancel, Accept or Reject

- **Online Payment Working**

In the user payment process via card with a payment gateway, the user selects the card as the payment method and enters card details during checkout. The payment gateway securely processes the transaction, transmitting information to the issuing bank for authorization. The issuing bank verifies the card and authorizes the transaction, with the payment gateway communicating the outcome to the online platform. The user receives confirmation, and the system records transaction details for reference. This use case ensures a secure and efficient payment experience, involving the coordinated efforts of the user, the payment gateway, and the issuing bank. As shown in Figure 3.4 .

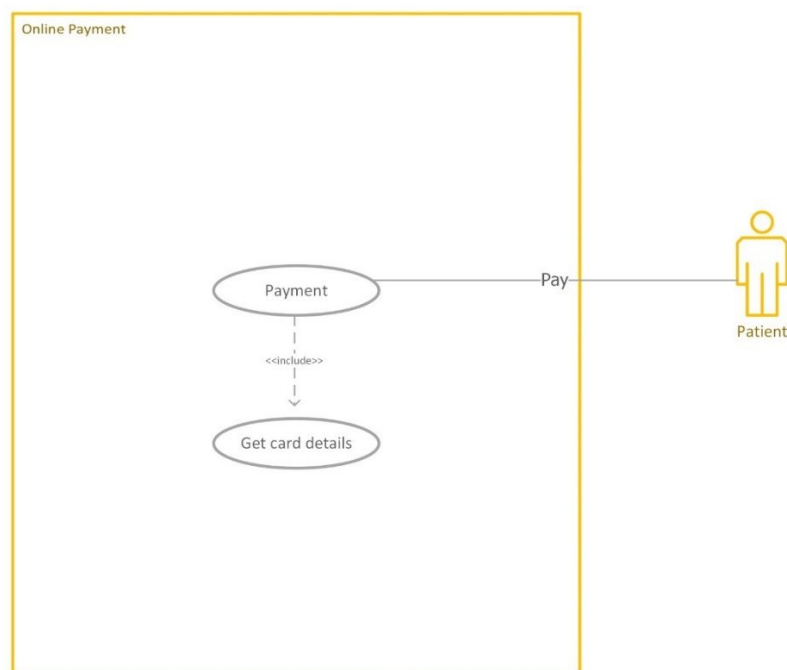


Figure 3.4 Use-Case Online Payment

- **Reviews**

The feature allowing doctors to receive, edit, and update reviews enhances transparency and trust within the healthcare platform. Patients can submit feedback, rating doctors based on various factors, contributing to an informed community. The ability to edit or update reviews provides flexibility for patients to refine their feedback. As shown in Figure 3.5 .

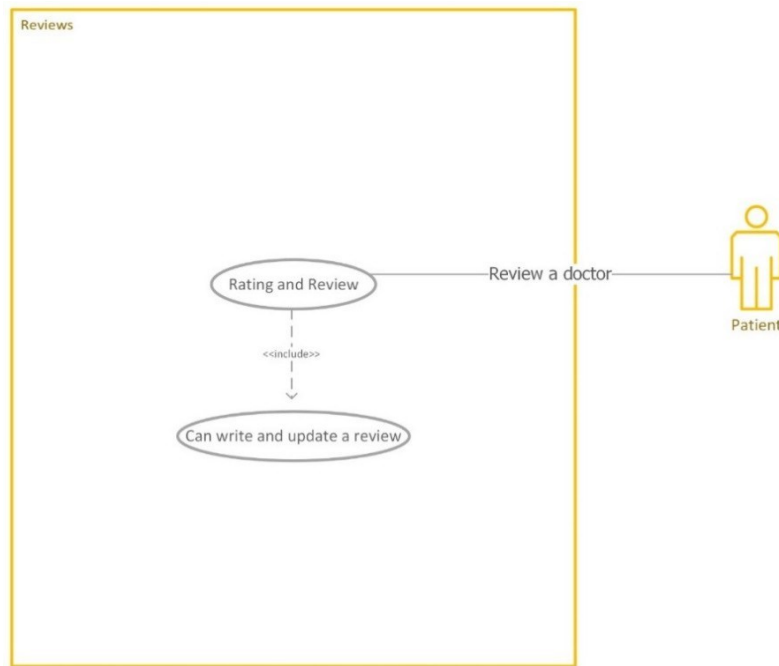


Figure 3.5 Use-case Review and Ratings of Doctor

- **Detailed Use Case**

A detailed use case is an exhaustive documentation of a specific interaction or functionality within a system. It includes key elements such as the use case name, primary actor, description, trigger, preconditions, postconditions, normal flow, alternative flows, exceptions, business rules and assumptions. This detailed breakdown provides a comprehensive understanding of the steps, conditions, and outcomes associated with a particular scenario. The use case documentation serves as a valuable tool for communication, aiding stakeholders, development teams, and other involved parties in understanding and implementing specific features within the system.

- **Detailed Use-case Patient and Doctor Registration**

The following Table 3.1 shows the details of Patient and Doctor Registration.

Table 3.1 Use-Case Patient and Doctor Registration

Use Case ID:	UC-1
UseCase Name:	Patient and Doctor Registration
Actors:	Customer

Description:	The E clinic System is an online platform that allows patients to access medical services, book appointments with doctors, view lab reports, and make online payments. It also provides a platform for doctors to manage their appointments, access patient information, and issue prescriptions.
Trigger:	• User clicks on the "Sign Up" button. • User clicks on the "Log In" button.
Preconditions:	• The user must have internet access. • The user must have a valid email address to sign up. • The user must have a registered account to log in.
Post conditions:	• Patients can book appointments physically at the clinic or online. • Patients can view their lab reports and prescriptions online. • Doctors can manage their appointments and access patient information. • Doctors can issue prescriptions for patients. • Patients can make online payments for their medical services.
Normal Flow:	• The Patient opens the E clinic application. • The E clinic application displays the login screen. • The Patient enters their username and password. • The E clinic application verifies the login credentials. • If the login is successful, the E clinic application displays the patient's dashboard. • The Patient selects the "Book Appointment" option. • The E clinic application presents a list of available doctors. • The Patient selects a doctor from the list. • The E clinic application shows the available time slots for the selected doctor. • The Patient chooses a convenient time slot for the appointment. • The E clinic application confirms the appointment and sends a notification to the Patient. • The Patient can view their upcoming appointments in the dashboard.

Alternative Flows:	• If the Patient does not have an account. • The Patient selects the "Sign Up" option. • The E clinic application displays the registration form. • The Patient enters their personal details, username, and password. • The E clinic application verifies the provided information. • If the registration is successful, the E clinic application creates a new account.
Exceptions:	• Invalid Login Credentials: The system should handle cases where users provide incorrect login credentials and display an error message. • Appointment Conflict: If a patient tries to book an appointment with a doctor who is already scheduled for another appointment at the same time, the system should inform the patient and suggest an alternative time slot. • Payment Failure: In case of payment failure during an online payment transaction, the system should notify the user and help with resolving the issue. • Prescription Access: Only authorized healthcare professionals should have access to view and prescribe medications. Unauthorized users should be prevented from accessing prescription-related functionality.
Business Rules:	• User Authentication: Users should be required to provide valid credentials (username and password) to access the system. • Doctor Availability: The system should display only those doctors who are available for appointments based on their schedule. • Appointment Confirmation: Once a patient successfully books an appointment, the system should send a confirmation notification to both the patient and the doctor. • Lab Report Access: Patients should be able to view their own lab reports, but only authorized healthcare professionals should be able to view reports of other patients. The seller must either accept, reject, or counter the offer within a specific timeframe.

	• The user must provide valid payment information to complete the purchase.
Assumptions:	• The E clinic system assumes that users have access to a reliable internet connection to access online features. • The system assumes that patients will provide accurate personal and medical information during the signup process. • It is assumed that doctors will maintain their schedules and update their availability in the system regularly. • The E clinic system assumes that online payment options are integrated with secure and trusted payment gateways.

• Detailed Use-case Patient Appointment Booking

The Following Table 3.2 shows the details of patient appointment booking.

Table 3.2 Use-Case Patient Appointment Booking

Use Case ID:	UC-2
UseCase Name:	Patient Appointment Booking
Actors:	Customer
Description:	The E clinic System is an online platform that allows patients to access medical services, book appointments with doctors, view lab reports, and make online payments. It also provides a platform for doctors to manage their appointments, access patient information, and issue prescriptions.
Triger:	• User books the Appointment. • User booked according to 'Date and Time.
Preconditions:	• The user must have internet access. • The user must have a valid email address to sign up. • The user must have a registered account to log in.
Post conditions:	• Patients can book appointments physically at the clinic or online. • Patients can view their lab reports and prescriptions online. • Doctors can manage their appointments and access patient information.

	• Doctors can issue prescriptions for patients. • Patients can make online payments for their medical services.
Normal Flow:	• The Patient opens the E clinic application. • The E clinic application displays the login screen. • The Patient enters their username and password. • The E clinic application verifies the login credentials. • If the login is successful, the E clinic application displays the patient's dashboard. • The Patient selects the "Book Appointment" option. • The E clinic application presents a list of available doctors. • The Patient selects a doctor from the list. • The E clinic application shows the available time slots for the selected doctor. • The Patient chooses a convenient time slot for the appointment. • The E clinic application confirms the appointment and sends a notification to the Patient. • The Patient can view their upcoming appointments in the dashboard.
Alternative Flows:	• If the Patient does not have an account: • The Patient selects the "Sign Up" option. • The E clinic application displays the registration form. • The Patient enters their personal details, username, and password. • The E clinic application verifies the provided information. • If the registration is successful, the E clinic application creates a new account and proceeds to step 4.
Exceptions:	• Invalid Login Credentials: The system should handle cases where users provide incorrect login credentials and display an error message. • Appointment Conflict: If a patient tries to book an appointment with a doctor who is already scheduled for another appointment at the same time, the system should inform the patient and suggest an alternative time slot.

	• Payment Failure: In case of payment failure during an online payment transaction, the system should notify the user and help with resolving the issue. • Prescription Access: Only authorized healthcare professionals should have access to view and prescribe medications. Unauthorized users should be prevented from accessing prescription-related functionality.
Business Rules:	• User Authentication: Users should be required to provide valid credentials (username and password) to access the system. • Doctor Availability: The system should display only those doctors who are available for appointments based on their schedule. • Appointment Confirmation: Once a patient successfully books an appointment, the system should send a confirmation notification to both the patient and the doctor. • Lab Report Access: Patients should be able to view their own lab reports, but only authorized healthcare professionals should be able to view reports of other patients. The seller must either accept, reject, or counter the offer within a specific timeframe. • The user must provide valid payment information to complete the purchase.
Assumptions:	• The E clinic system assumes that users have access to a reliable internet connection to access online features. • The system assumes that patients will provide accurate personal and medical information during the signup process. • It is assumed that doctors will maintain their schedules and update their availability in the system regularly. • The E clinic system assumes that online payment options are integrated with secure and trusted payment gateways.

- **Detailed Use-case Patient Appointment Cancel, Accept or Reject**

The following Table 3.3 shows the details of use-case Patient Appointment Cancel, Accept or Reject

Table 3.3 Use-Case Patient Appointment Cancel, Accept or Reject

Use Case ID:	UC-3
UseCase Name:	Patient Appointment
Actor:	Customer
Description:	The E clinic System is an online platform that allows patients to access medical services, book appointments with doctors, view lab reports, and make online payments. It also provides a platform for doctors to manage their appointments, access patient information, and issue prescriptions.
Triger:	• There is a time of 24 hours of cancellation or confirmation. • Either 'booked' or 'cancelled'.
Preconditions:	• The user must have internet access. • The user must have a valid email address to sign up. • The user must have a registered account to log in.
Post conditions:	• Patients can book appointments physically at the clinic or online. • Patients can view their lab reports and prescriptions online. • Doctors can manage their appointments and access patient information. • Doctors can issue prescriptions for patients. • Patients can make online payments for their medical services.
Normal Flow:	• The Patient opens the E clinic application. • The E clinic application displays the login screen. • The Patient enters their username and password. • The E clinic application verifies the login credentials. • If the login is successful, the E clinic application displays the patient's dashboard. • The Patient selects the "Book Appointment" option. • The E clinic application presents a list of available doctors. • The Patient selects a doctor from the list. • The E clinic application shows the available time slots for the selected doctor. • The Patient chooses a convenient time slot for the appointment.

	• The E clinic application confirms the appointment and sends a notification to the Patient. • The Patient can view their upcoming appointments in the dashboard.
Alternative Flows:	• If the Patient does not have an account: • The Patient selects the "Sign Up" option. • The E clinic application displays the registration form. • The Patient enters their personal details, username, and password. • The E clinic application verifies the provided information. • If the registration is successful, the E clinic application creates a new account and proceeds to step 4.
Exceptions:	• Invalid Login Credentials: The system should handle cases where users provide incorrect login credentials and display an error message. • Appointment Conflict: If a patient tries to book an appointment with a doctor who is already scheduled for another appointment at the same time, the system should inform the patient and suggest an alternative time slot. • Payment Failure: In case of payment failure during an online payment transaction, the system should notify the user and help with resolving the issue. • Prescription Access: Only authorized healthcare professionals should have access to view and prescribe medications. Unauthorized users should be prevented from accessing prescription-related functionality.
Business Rules:	• User Authentication: Users should be required to provide valid credentials (username and password) to access the system. • Doctor Availability: The system should display only those doctors who are available for appointments based on their schedule.

	• Appointment Confirmation: Once a patient successfully books an appointment, the system should send a confirmation notification to both the patient and the doctor. • Lab Report Access: Patients should be able to view their own lab reports, but only authorized healthcare professionals should be able to view reports of other patients. The seller must either accept, reject, or counter the offer within a specific timeframe. • The user must provide valid payment information to complete the purchase.
Assumptions:	• The E clinic system assumes that users have access to a reliable internet connection to access online features. • The system assumes that patients will provide accurate personal and medical information during the signup process. • It is assumed that doctors will maintain their schedules and update their availability in the system regularly. • The E clinic system assumes that online payment options are integrated with secure and trusted payment gateways.

• **Detailed Use-case Online Payment:**

The following Table 3.4 shows the details of Use-case Online Payment.

Table 3.4 Use-Case Online Payment

Use Case ID:	UC-4
UseCase Name:	Online Payment
Actors:	Customer
Description:	The E clinic System is an online platform that allows patients to access medical services, book appointments with doctors, view lab reports, and make online payments. It also provides a platform for doctors to manage their appointments, access patient information, and issue prescriptions.

Trigger:	• Users pay his payment by credit card.
Preconditions:	• The user must have internet access. • The user must have a valid email address to sign up. • The user must have a registered account to log in.
Post conditions:	• Patients can book appointments physically at the clinic or online. • Patients can view their lab reports and prescriptions online. • Doctors can manage their appointments and access patient information. • Doctors can issue prescriptions for patients. • Patients can make online payments for their medical services.
Normal Flow:	• The Patient opens the E clinic application. • The E clinic application displays the login screen. • The Patient enters their username and password. • The E clinic application verifies the login credentials. • If the login is successful, the E clinic application displays the patient's dashboard. • The Patient selects the "Book Appointment" option. • The E clinic application presents a list of available doctors. • The Patient selects a doctor from the list. • The E clinic application shows the available time slots for the selected doctor. • The Patient chooses a convenient time slot for the appointment. • The E clinic application confirms the appointment and sends a notification to the Patient. • The Patient can view their upcoming appointments in the dashboard.
Alternative Flows:	• If the Patient does not have an account: • The Patient selects the "Sign Up" option. • The E clinic application displays the registration form. • The Patient enters their personal details, username, and password. • The E clinic application verifies the provided information.

	• If the registration is successful, the E clinic application creates a new account and proceeds to step 4.
Exceptions:	• Invalid Login Credentials: The system should handle cases where users provide incorrect login credentials and display an error message. • Appointment Conflict: If a patient tries to book an appointment with a doctor who is already scheduled for another appointment at the same time, the system should inform the patient and suggest an alternative time slot. • Payment Failure: In case of payment failure during an online payment transaction, the system should notify the user and provide assistance for resolving the issue. • Prescription Access: Only authorized healthcare professionals should have access to view and prescribe medications. Unauthorized users should be prevented from accessing prescription-related functionality.
Business Rules:	• User Authentication: Users should be required to provide valid credentials (username and password) to access the system. • Doctor Availability: The system should display only those doctors who are available for appointments based on their schedule. • Appointment Confirmation: Once a patient successfully books an appointment, the system should send a confirmation notification to both the patient and the doctor. • Lab Report Access: Patients should be able to view their own lab reports, but only authorized healthcare professionals should be able to view reports of other patients. The seller must either accept, reject, or counter the offer within a specific timeframe. • The user must provide valid payment information to complete the purchase.

Assumptions:	• The E clinic system assumes that users have access to a reliable internet connection to access online features. • The system assumes that patients will provide accurate personal and medical information during the signup process. • It is assumed that doctors will maintain their schedules and update their availability in the system regularly. • The E clinic system assumes that online payment options are integrated with secure and trusted payment gateways.
---------------------	--

• **Detailed Use-case Review and Ratings of Doctor:**

The following Table 3.5 shows the details of Use-case Review and Ratings.

Table 3.5 Use-Case Review and Ratings of Doctor

Use Case ID:	UC-5
UseCase Name:	Use-case Review and Ratings of Doctor
Actors:	Customer
Description:	The E clinic System is an online platform that allows patients to access medical services, book appointments with doctors, view lab reports, and make online payments. It also provides a platform for doctors to manage their appointments, access patient information, and issue prescriptions.
Trigger:	• Users give reviews about his/her experience. • Users update the reviews
Preconditions:	• The user must have internet access. • The user must have a valid email address to sign up. • The user must have a registered account to log in.
Post conditions:	• Patients can book appointments physically at the clinic or online. • Patients can view their lab reports and prescriptions online. • Doctors can manage their appointments and access patient information. • Doctors can issue prescriptions for patients. • Patients can make online payments for their medical services.

Normal Flow:	• The Patient opens the E clinic application. • The E clinic application displays the login screen. • The Patient enters their username and password. • The E clinic application verifies the login credentials. • If the login is successful, the E clinic application displays the patient's dashboard. • The Patient selects the "Book Appointment" option. • The E clinic application presents a list of available doctors. • The Patient selects a doctor from the list. • The E clinic application shows the available time slots for the selected doctor. • The Patient chooses a convenient time slot for the appointment. • The E clinic application confirms the appointment and sends a notification to the Patient. • The Patient can view their upcoming appointments in the dashboard.
Alternative Flows:	• If the Patient does not have an account: • The Patient selects the "Sign Up" option. • The E clinic application displays the registration form. • The Patient enters their personal details, username, and password. • The E clinic application verifies the provided information. • If the registration is successful, the E clinic application creates a new account and proceeds to step 4.
Exceptions:	• Invalid Login Credentials: The system should handle cases where users provide incorrect login credentials and display an error message. • Appointment Conflict: If a patient tries to book an appointment with a doctor who is already scheduled for another appointment at the same time, the system should inform the patient and suggest an alternative time slot.

	• Payment Failure: In case of payment failure during an online payment transaction, the system should notify the user and help with resolving the issue. • Prescription Access: Only authorized healthcare professionals should have access to view and prescribe medications. Unauthorized users should be prevented from accessing prescription-related functionality.
Business Rules:	• User Authentication: Users should be required to provide valid credentials (username and password) to access the system. • Doctor Availability: The system should display only those doctors who are available for appointments based on their schedule. • Appointment Confirmation: Once a patient successfully books an appointment, the system should send a confirmation notification to both the patient and the doctor. • Lab Report Access: Patients should be able to view their own lab reports, but only authorized healthcare professionals should be able to view reports of other patients. The seller must either accept, reject, or counter the offer within a specific timeframe. • The user must provide valid payment information to complete the purchase.
Assumptions:	• The E clinic system assumes that users have access to a reliable internet connection to access online features. • The system assumes that patients will provide accurate personal and medical information during the signup process. • It is assumed that doctors will maintain their schedules and update their availability in the system regularly. • The E clinic system assumes that online payment options are integrated with secure and trusted payment gateways.

3.2 Functional Requirements

These requirements describe what the software is supposed to do and how it should perform in various situations. Functional requirements are typically documented during the early stages of the software development life cycle and serve as a foundation for design and implementation.

1. User Authentication

Login: Users should be able to log in to their accounts using their credentials.

Signup: New users should be able to create an account by providing the required information.

For authentication used JWT protocol.

2. User Roles

Patients: Patients should have access to features like appointment booking, checking doctors, prescription history, lab reports, and online payment.

3. Manage Doctors

Doctors should have access to features like managing appointments, checking patient details, prescribing medications, and reviewing lab reports.

4. Appointment Booking

Physical Booking: Patients should be able to book appointments by visiting the clinic in person.

Online Booking: Patients should be able to book appointments through the E clinic platform, selecting the desired doctor, date, and time.

5. Doctor Availability

Check Doctors: Patients should be able to view the list of available doctors, along with their specialties, schedules, and ratings.

6. Prescription Management

Prescription Button: Doctors should have the option to generate prescriptions for patients, including medications, dosage instructions, and duration.

Prescription History: Patients should be able to view their prescription history, including details of past medications prescribed by doctors.

7. Lab Reports

Lab Reports Upload: Patients should have the ability to upload lab reports to their profiles.

Lab Reports Review: Doctors should be able to access and review the lab reports uploaded by patients.

8. Online Payment

Payment Integration: Patients should be able to make online payments for services like consultation fees, prescription charges, or lab tests.

9. Payment Gateway

The E clinic platform should integrate with a secure and reliable payment gateway for processing online transactions.

These modules cover the essential functional requirements of an E clinic system, including user authentication, user roles, appointment booking (both physical and online), doctor availability, prescription management, lab reports, and online payment options. Additional modules may be added based on specific requirements or customization needs.

3.2.1 Doctor Module

The Doctor modules performs the following functionalities:

1. **Profile Management:** Allows doctors to update their professional information, contact details, and availability.
2. **Appointment Management:** Enables doctors to view and manage their appointment schedules, accept, or reschedule appointments, and block time slots when they are unavailable.
3. **Prescription Writing:** Allows doctors to create and manage prescriptions for their patients, including medication details and dosage instructions.
4. **Lab Report Access:** Enables doctors to view and analyze lab reports and test results of their patients.

3.2.2 Patient Module

The Patient module performs the following functionalities:

1. **Profile Management:** Allows patients to update their personal information, contact details, and medical history.
2. **Doctor Search:** Enables patients to search for doctors based on various criteria such as specialization, location, and availability.
3. **Appointment Booking:** Allows patients to schedule appointments with their preferred doctors, either physically or online.
4. **Prescription Management:** Provides a platform for patients to access and manage their prescriptions, view medication details, and request refills.
5. **Lab Reports:** Allows patients to view their lab reports and test results.

6. **Online Payment:** Provides a secure payment gateway for patients to make online payments for appointments, prescriptions, and other services.

3.2.3 User Authentication Module

The User Authentication module performs the following functionalities:

1. **User Authentication Module:**
2. **Login:** Allows users (patients and doctors) to log in to their accounts.
3. **Signup:** Enables new users to create accounts and register.

3.2.4 Admin Module

The admin module performs the following functionalities:

1. **User Management:** Enables the administrator to manage user accounts, including patient and doctor registrations.
2. **Doctor Management:** Allows the administrator to add, modify, or remove doctor profiles from the system, including their specialties and availability.
3. **Appointment Tracking:** Provides an overview of all appointments, allowing the administrator to monitor and manage the appointment scheduling process.
4. **Payment Management:** Enables the administrator to manage online payments, track transactions, and generate reports.

3.3 Non-Functional Requirements

Non-Functional Requirements are the requirements that specify criteria that can be used to judge the operation of a system. Those constraints under which the system will be operated are called non-functional requirements. For example, language run time environment, operating environment performance requirements, usability requirements etc. These are all those requirements which do not belong to functional requirements, but they affect the system overall. We can say some extra conditions requirements that are not included in the use cases. These are usually called non-functional requirements; some of these are given below.

Following are the non-functional requirements:

- Security
- Performance
- Reliability
- Usability

3.3.1 Security

Security is a crucial non-functional requirement that ensures the protection of the system and its data. Here are key considerations for incorporating security into non-functional requirements:

- a) The system should ensure secure communication by using encryption protocols (e.g., AES) for password encryption.
- b) User authentication should be implemented to verify the identity of patients, doctors, and administrative staff.
- c) Access controls should be enforced to restrict unauthorized access to sensitive information and system functionalities.

3.3.2 Performance

Performance is a critical non-functional requirement in software development, focusing on how well a system performs in terms of speed, responsiveness, scalability, and efficiency. Ensuring optimal performance is essential for providing a satisfactory user experience and meeting the demands of various workloads.

- a) The system should have fast response times for login, signup, appointment booking, and other critical operations to ensure a smooth user experience.
- b) It should be capable of handling many concurrent users without significant degradation in performance.
- c) The system should be scalable to accommodate future growth and increased user traffic.

3.3.3 Reliability

Reliability is a crucial non-functional requirement in software development, focusing on the ability of a system to consistently perform its intended functions accurately and dependably over time. Ensuring reliability is essential for building trust among users and stakeholders.

- a) The system should be available and accessible to users 24/7, with minimal downtime for maintenance or upgrades.
- b) It should have backup and disaster recovery mechanisms to prevent data loss and ensure quick system restoration in case of failures or emergencies.
- c) The system should have appropriate monitoring and alerting mechanisms to promptly identify and resolve issues, ensuring high system availability.

3.3.4 Usability

Usability is a vital non-functional requirement in software development, focusing on how easily and effectively users can interact with and navigate through a system. It emphasizes providing a positive and efficient user experience.

- a)** The user interface should be intuitive, user-friendly, and responsive across different devices and screen sizes.
- b)** Clear and concise instructions should be provided to guide users through the registration process, appointment booking, and other system functionalities.
- c)** The system should support multiple languages, considering the diverse user base it may serve.

Chapter 4

Design and Architecture

4 Design and Architecture

Design is a phase within the broader architectural process. The architecture sets the framework, and design fills in the details. Software design refers to the process of defining the architecture, components, modules, interfaces, and data for a system to satisfy specified requirements.

4.1 System Architecture

The structure of the system explains the working of the system and describes the components of the project, their relationships, and how they interact with each other and with the architecture. As Shown in Figure 4.1.

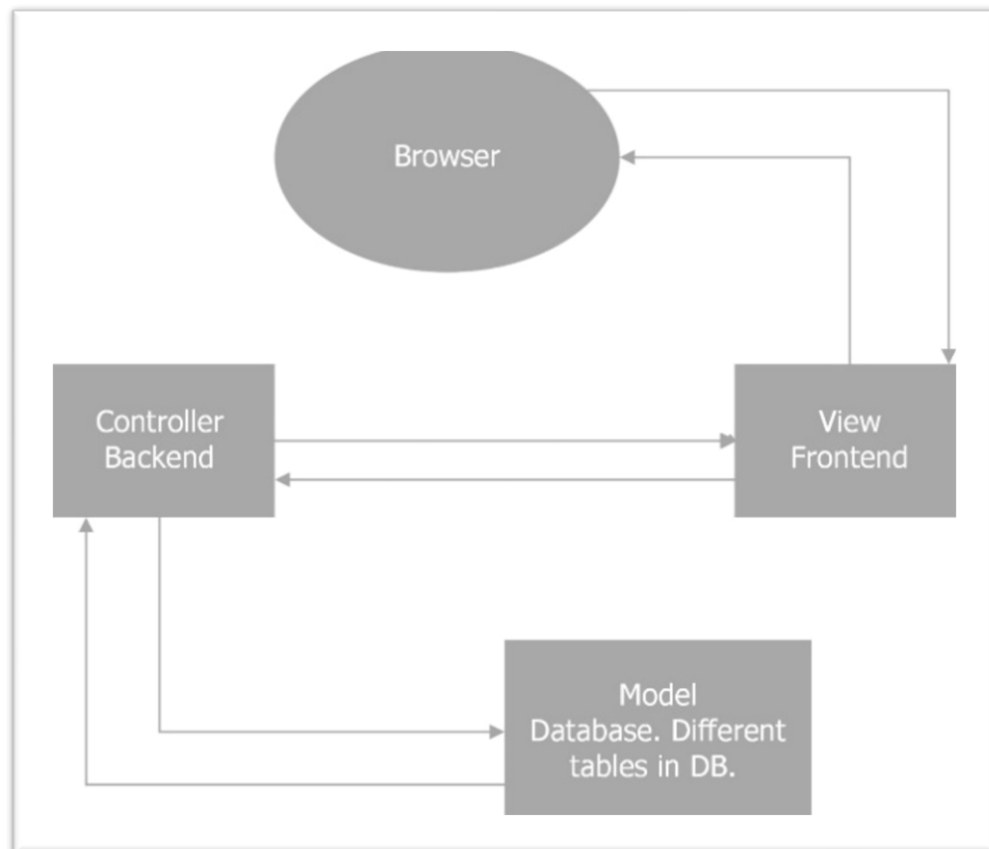


Figure 4.1 System Architecture of E clinic

4.2 Data Representation

The following Table 4.1 shows the data representation as follows.

Table 4.1 Data Representation of E clinic

Attributes	Data Type	Description
PatientID	Integer	Unique identifier for the patient
Name	String	User's full name
Email	String	User's email address
Password	String	User's password (encrypted)
User Type	String	Type of user (patient/doctor)
Reason for Appointment	String	Gives the reason of appointment
Availability	String	Doctor's availability hours

Attributes	Data Type	Description
Medications Prescribed	String	List of medications with dosage and instructions
SessionID	Integer	Unique identifier for the session
PatientUserID	Integer	UserID of the patient participating in the session
DoctorID	Integer	UserID of the doctor participating in the session
Registration Date	Date Time	Date time of the registration
Payment ID (Unique identifier)	integer	ID of payment
Payment Method	String	Payment method of the patient

4.3 Process Flow

A process flow in software development refers to the sequence of steps or activities that outline the path a project or a specific task follows from initiation to completion. It provides a visual representation of the series of actions, decisions, and interactions among individuals or components involved in the software development lifecycle. Process flows are instrumental in understanding, communicating, and managing the various stages of software development.

1. Patient Registration

Patients register on the app by providing their details and creating an account.

2. Doctors Appointment

Patients take the appointment from the doctor with specific time.

3. Doctor Search

Patients search for doctors based on their technology interests and expertise.

4. Prescription

The doctor gives the prescription according to the patient's health issue.

5. Appointment Confirm

Patients confirm his appointment in between 24 hours' time.

- **Diagram**

The Figure 4.2 Shows the complete process flow of E clinic.

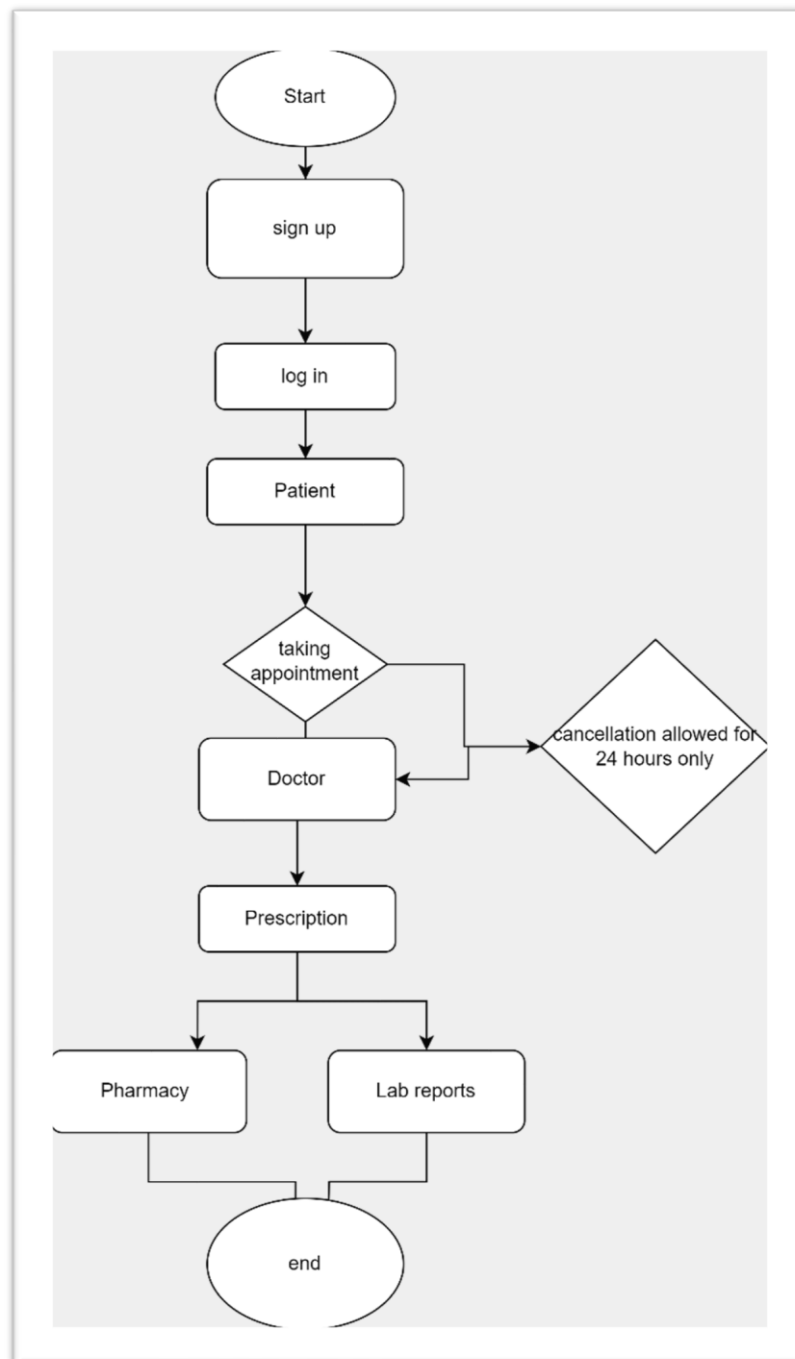


Figure 4.2 Complete Process Flow of E clinic

4.4 Design Models

The design model of the E clinic is as follows.

4.4.1 Class diagram

The class diagram Figure 4.3 of the e clinic consists of classes such as patient, doctor, appointment, location. Class diagram is a type of UML (Unified Modelling Language) diagram used in software engineering to visualize the structure of a system by showing the classes, their attributes, methods, and the relationships between them.

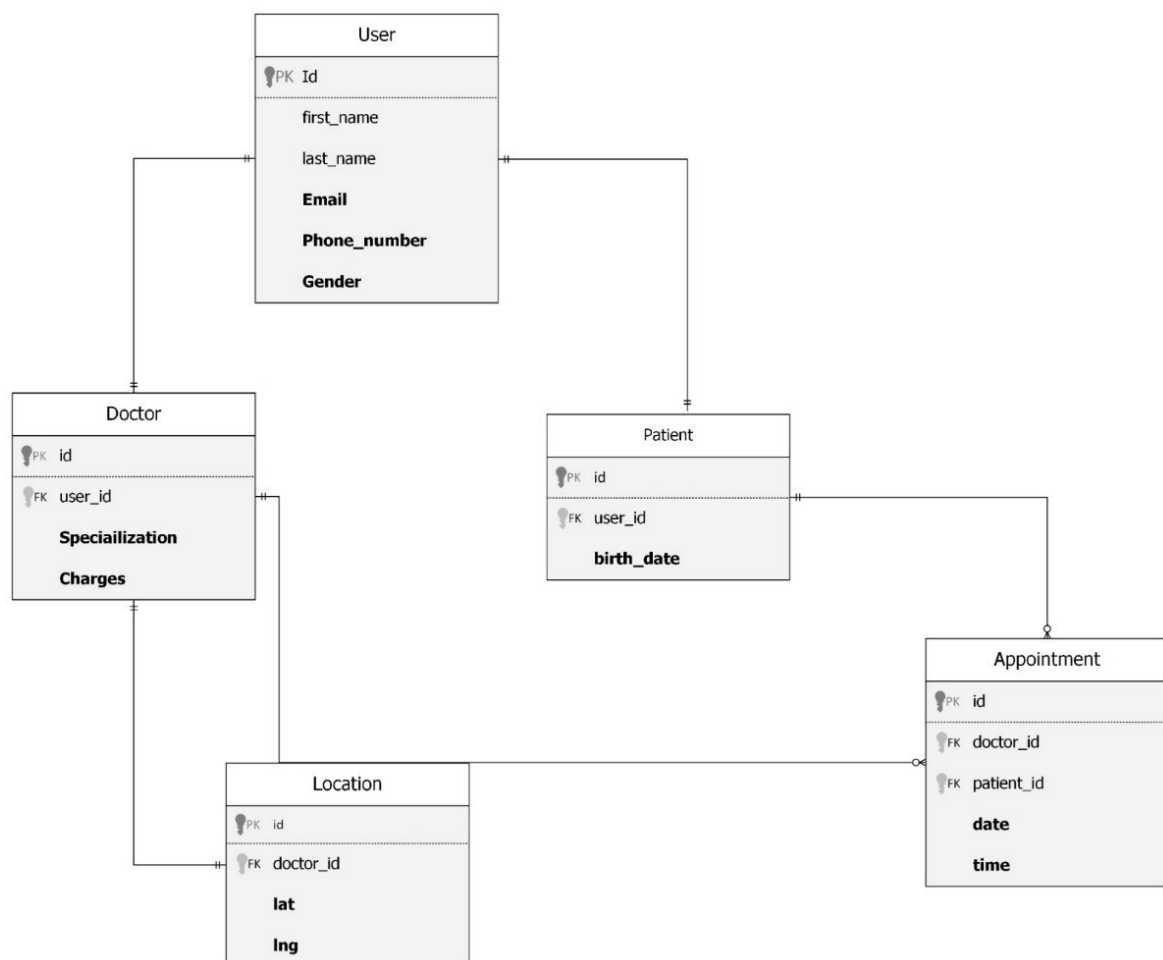


Figure 4.3 Class Diagram of E clinic

4.1.2 Sequence Diagram

The sequence diagram for the mentorship app illustrates the interactions between the patient, doctor, and other features. It shows the sequence of actions, such as appointment, doctor and time, prescription, and lab report, highlighting the flow of communication and processes within the app's functionality. As Shown in Figure 4.4

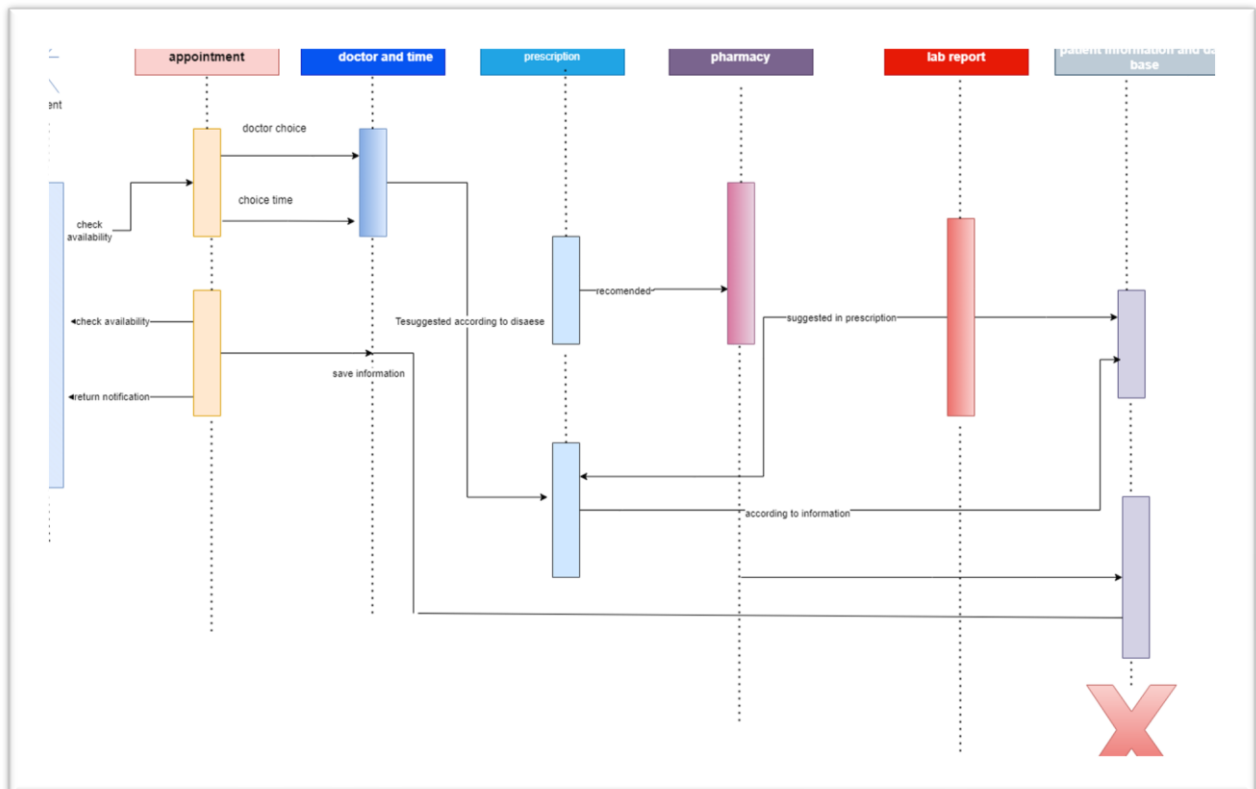


Figure 4.4 Sequence Diagram of E clinic

Chapter 5

Implementation

5 Implementation

The following is a high-level description of the implementation and algorithm for the application using microservices architecture:

1. Authentication Service

Implement an authentication service that handles user login and logout. Verify user credentials and determine their role (doctor or patient).

2. Patient Account

Patients should be able to register for an account on the platform. This typically involves providing personal information like name, email address, phone number, and a password.

Implement email verification to ensure that the provided email address is valid.

Store user registration data in a secure database. Patients should be able to log in to their accounts using their email address and password.

3. Doctor Account

Create a doctor account where doctors can receive messages from patients and respond to them.

Enable doctors to view and reply to messages received from patients. Doctors should be able to log in using their registered email address and a secure password.

➤ Appointment Booking

The following are the flow of appointment booking.

- **Appointment Creation**

Allow patients to request appointments with doctors through the platform.

Provide a user-friendly interface for patients to select a preferred date, time, and doctor or specialty. Implement validation checks to prevent overlapping or conflicting appointments.

- **Appointment Availability**

Doctors should set their availability schedules, specifying when they can accept appointments.

Implement a calendar system that displays available time slots for each doctor. Ensure real-time updates to reflect any changes in doctor availability.

- **Appointment Confirmation**

Doctors should receive notifications of pending appointment requests. Doctors can either manually confirm or decline appointments.

- **Database Integration**

Integrate a database to store user information, messages, and other relevant data.

Implement appropriate database models and relationships.

- **Scalability and Resilience**

Implement scalability measures to handle many users and messages. Ensure fault tolerance and resilience by implementing backup mechanisms and error handling.

- **Testing and Deployment**

Conduct thorough testing to ensure the functionality, performance, and reliability of the application.

Testing: Perform comprehensive unit, integration, and end-to-end testing to identify and fix any bugs or performance issues. This ensures each microservice functions as expected both individually and as part of the overall system. Automated testing can also be implemented for faster and more reliable releases.

Deployment: Deploy the microservices on a scalable infrastructure, such as AWS, Azure, or Kubernetes, ensuring proper load balancing, fault tolerance, and real-time monitoring. This setup helps maintain high availability and seamless user experiences even under heavy loads.

5.1 User Interface

The system design will be simple and easy to use. Users can easily navigate to the buttons. There will be a user-friendly GUI. Patients can easily select the buttons to start chatting with doctors. The design will include clear icons and labels to guide users effectively. It will also ensure quick response times for a smooth user experience.

i. Login/Signup Page

Figure 5.1 shows the login page for the doctor or patient. Here a user will provide his/her username and password to log in.

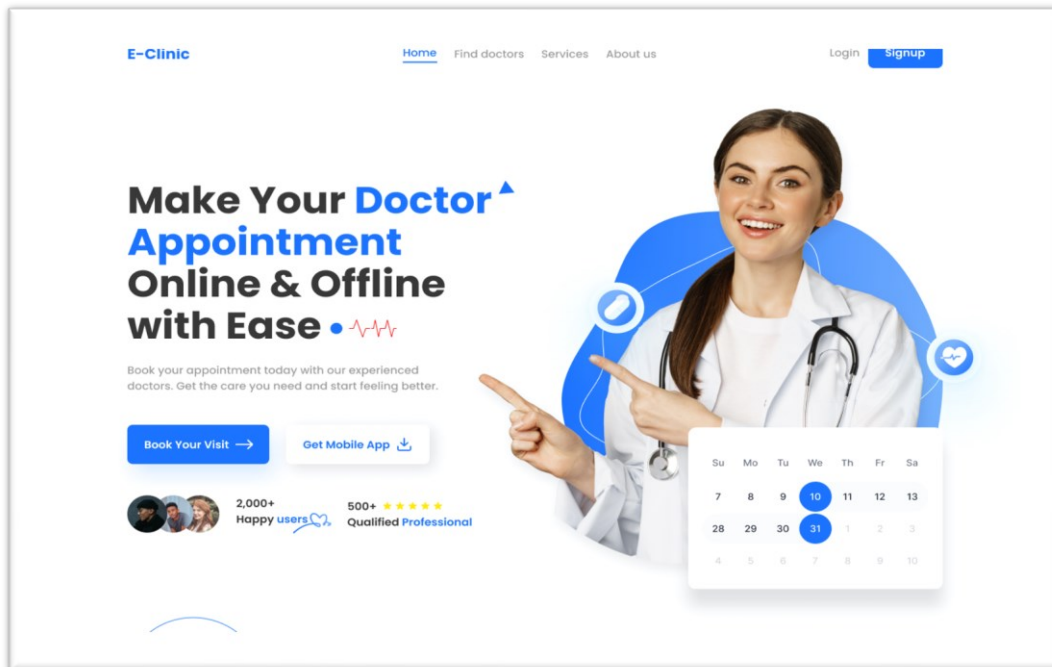


Figure 5.1 Login/Signup of E clinic

ii. Features

Figure 5.2 shows the page where we can search the doctor and find by writing the name of the specific mentor.

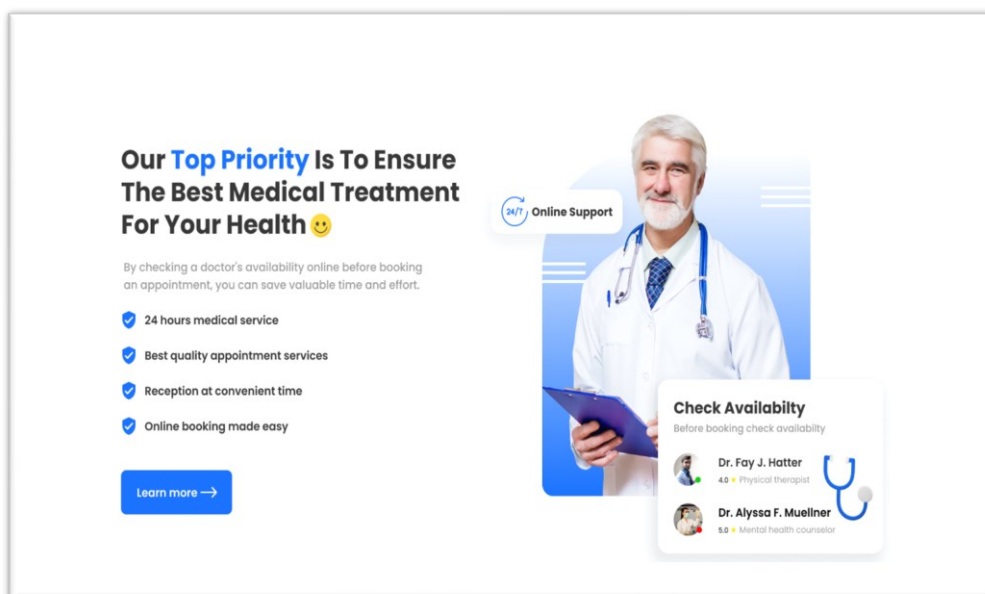


Figure 5.2 Feature of E clinic

iii. Dashboard

Figure 5.3 shows the different options where a patient connects with the doctor. This is basically the different features for the patient.

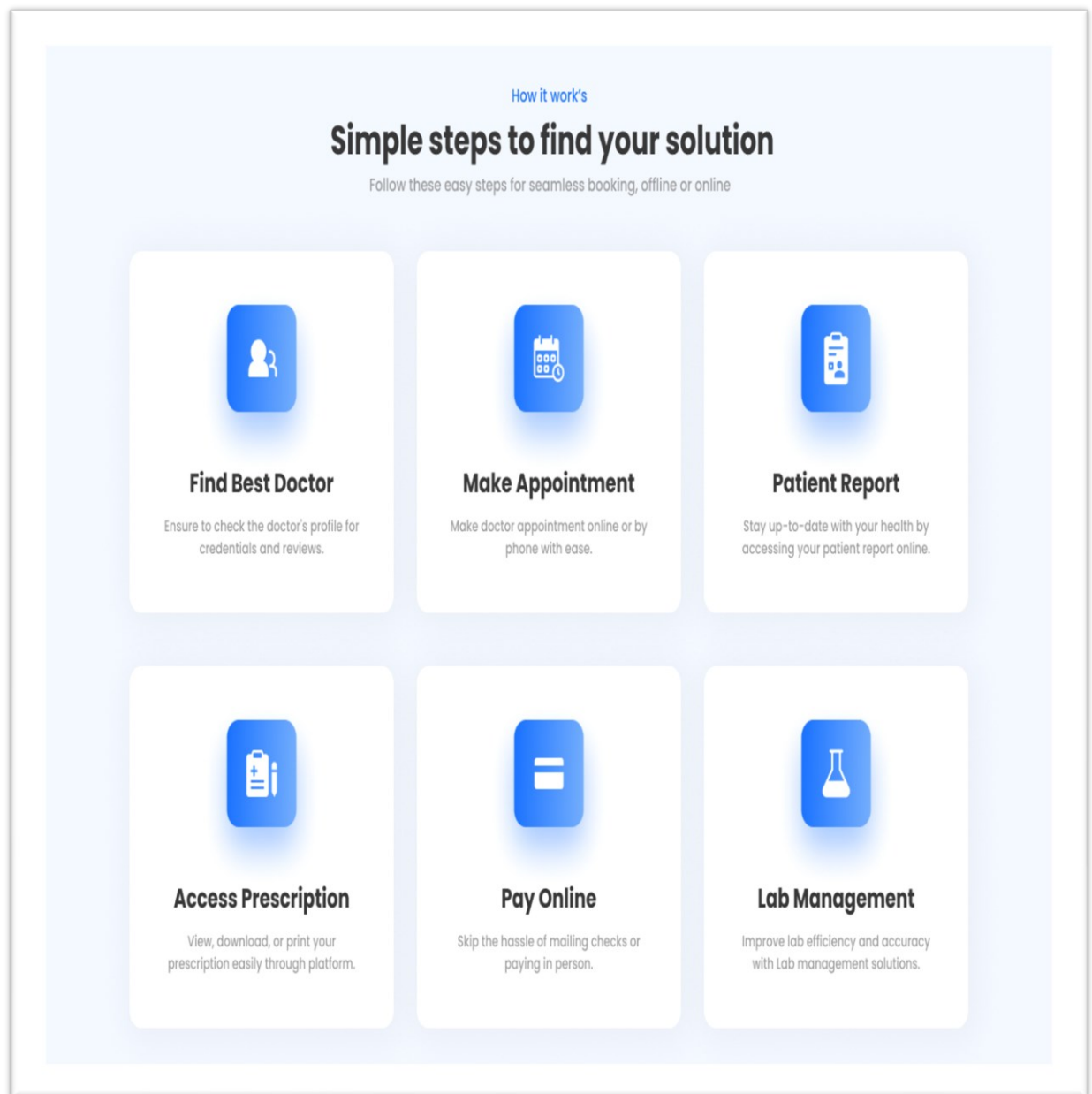


Figure 5.3 Dashboard for Patients

iv. Doctors

Figure 5.4 shows the page where a patient can check and visit different doctors.

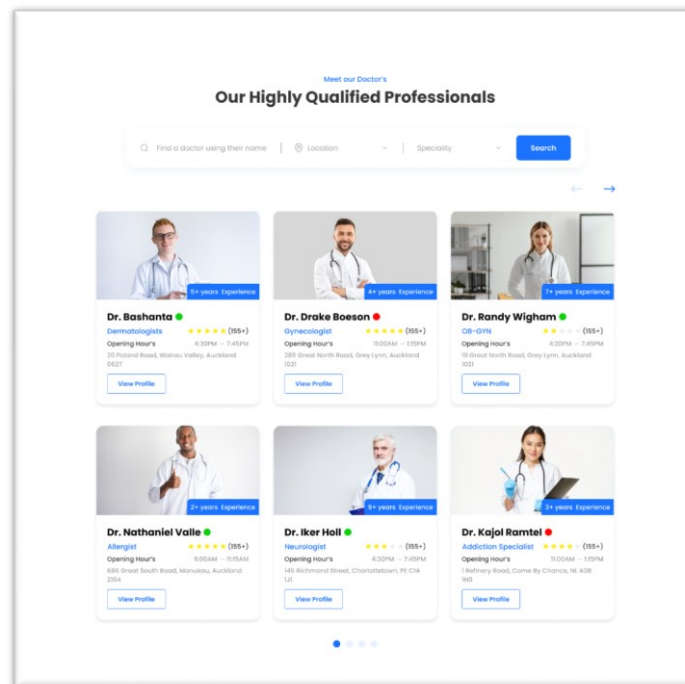


Figure 5.4 Doctors Page of E clinic

v. About

Figure 5.4 shows the page where a patient can check and visit different doctors.

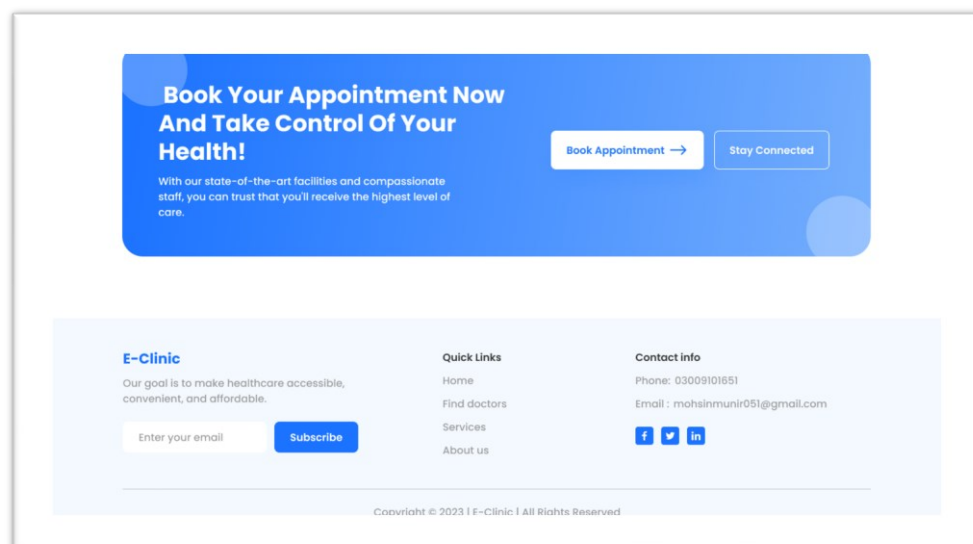


Figure 5.5 About Us Page of E clinic

Chapter 6

Testing and Evaluation

6 Testing and Evaluation

The testing and evaluation phase is a crucial component of the project, ensuring that the developed application meets the desired requirements and is reliable and functional. This chapter outlines the various types of testing that will be conducted during the project, including manual testing, system testing, unit testing and functional testing.

6.1 Manual Testing

Manual testing is the process of verifying the functionality of the app manually by the testers. This process is essential to ensure that the app and site meets the user requirements and performs as expected. The following Table 6.1 shows the test cases and results for manual testing. As Shown in Table

Table 6.1 Manual Testing of E clinic

Test Cases	Expected Result	Actual Result	Pass/Fail
App Launch	Successful	Successful	Pass
Patient Login	Successful	Successful	Pass
Patient Registration	Successful	Successful	Pass
App and site Performance and Response	Smooth	Smooth	Pass
App and site Usability and User Friendliness	High	High	Pass

6.2 System Testing

System testing is the process of verifying that the application functions as intended in the overall system environment. The following Table 6.2 the test cases and results for system testing:

Table 6.2 System Testing of E clinic

Test Cases	Expected Result	Actual Result	Pass/Fail
Compatibility with Device	Compatible	Compatible	Pass
Data Security	Secure	Secure	Pass
Data Backup	Reliable	Reliable	Pass
User Management	Effective	Effective	Pass
App Response Time	Acceptable	Acceptable	Pass

Chapter 7

Conclusion and Future Work

7. Conclusion

In conclusion, the E clinic system described in the provided information is a comprehensive healthcare platform that aims to enhance patient engagement and streamline healthcare services through the integration of patient and doctor modules. This system is designed to improve healthcare outcomes by facilitating better communication, access to medical records, and appointment management.

Patients can access a range of features, such as viewing messages from doctors, checking doctor availability, and communicating securely with healthcare providers. They can also browse and select doctors from various professions, leave reviews based on their experiences, and access their appointment history.

Doctors can sign up and log in to their accounts, providing their personal information, professional details, and experience. They can manage appointments, including accepting or canceling upcoming appointments. The system also offers password recovery options and access to payment information.

7.1 Future Work

Implementing machine learning algorithms to enhance your E clinic system's ability to diagnose diseases from different symptoms and provide general precautions and remedies is a promising avenue for future work. Here are some potential steps and considerations for incorporating machine learning into your healthcare platform:

Symptom Checker and Diagnostic Assistant:

Develop a symptom checker module that allows patients to input their symptoms, and then uses machine learning algorithms to provide potential diagnoses or recommend necessary medical tests. This can help users get a preliminary understanding of their condition before consulting a doctor.

Disease Prediction and Early Detection:

Use historical patient data and machine learning models to predict the likelihood of developing certain diseases based on genetics, lifestyle, and medical history. Early detection and prevention can improve patient outcomes.

Personalized Treatment Recommendations:

Implement algorithms that analyse a patient's medical history, test results, and current condition to generate personalized treatment recommendations. This can include medication dosage suggestions, lifestyle changes, and therapy options.

Medication Adherence Monitoring:

Develop a system that uses machine learning to monitor patient adherence to prescribed medications. Choosing the right machine learning technique or combination of techniques depends on the specific requirements of our e clinic site and app and the available data. Collaborating with domain experts and conducting rigorous testing and validation are essential steps in the development and deployment of machine learning models for disease diagnosis.

8. References

- Hashami, M. (2020). Healthcare Systems & its Challenges in Pakistan. *International Journal of Social Sciences*, 09, 23. Retrieved 03 2023, from https://www.researchgate.net/publication/341937017_Healthcare_Systems_its_Challenges_in_Pakistan
- Imam, E. (2016, Jan). *Marham*. Retrieved March 2023, from <https://www.marham.pk/>
- Kidwai, F. (2013, Nov). *Dawaai*. (CB Insights) Retrieved March 2023, from <https://dawaai.pk/>
- Muhammad Q, E. H. (2023). Healthcare in Pakistan: Navigating Challenges and Building Brighter Future. *Cureus*, 15(6), 2. Retrieved 03 2023, from <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10332330/>
- Zuberi, A. (2017, March). *Oladoc*. Retrieved March 2023, from <https://oladoc.com/>
- Khan, M. A., & Khan, N. (2021). *Telemedicine in Pakistan: Challenges, Prospects, and Future*. *Journal of the Pakistan Medical Association*, 71(12), 2839–2841. Retrieved March 2023, from <https://jpma.org.pk/article-details/11255>
- Ministry of National Health Services, Regulations & Coordination. (2021). *National Health Vision Pakistan 2019*. Government of Pakistan. Retrieved March 2021, from <http://www.nhsrcc.gov.pk/>